

to ensure integrity is to perform a checksum on the information being transmitted and to transmit the checksum in an encrypted form as well. The information is received on the other end and again checksummed. The transmitted checksum is decrypted and compared with the previous checksum. If the checksums agree, the information is most likely unaltered. The problem with this scheme is that the checksum of the original message can be known and another message with the same checksum can be generated and sent instead of the original one. This problem can be overcome by using a public-key cryptosystem. After generating the public-key/private-key pair, if we throw away the private-key and use only the public-key to encrypt the checksum, the checksum becomes impossible to decrypt. In order to verify the checksum, we generate a new checksum for the received information, encrypt it using the public-key and match it with the encrypted checksum. This is also known as a one-way function as it is hard to invert.

Integrity in storage

Integrity in storage had been ensured by access control systems with lock and keys and other guards to prevent unauthorized access to stored data. The existence of computer viruses has changed the scenario and the need of integrity against intentional attack has become a problem of epic proportions. Cryptographic checksums to ascertain validity of stored data are of help here. As in the case of transmission, a cryptographic checksum is produced and compared to the expected value. However, storage media are more vulnerable to attacks than transmission channels due to longer exposure and larger volumes of information.

Authentication of identity

Authentication is the process of verifying if the user has enough authority for data access. Simple passwords are used to identify someone. You must also have seen in classic gangster movies, the exchange of keywords to prove identity. Cryptography is similar to the practice of providing passwords for identity authentication. Modern systems use cryptographic transforms in conjunction with other characteristics of individuals to provide more reliable and efficient authentication of identity. Many systems allow passwords to be stored in an encrypted form, with read access available to all programs which may use them. Since passwords are not stored as plaintext, an accidental leak of data doesn't compromise the system's security.